

Analysis of Voltage Dip Source Location Methods

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Abstract— The voltage dip is one of the prominent disturbances in power systems related to power quality. Therefore it is very important to investigate the methods which allow to determine voltage dip direction finder. In this work selected voltage dip source location methods are discussed and investigated. The analysis are performed in Matlab Simulink environment. The methods are tested using different kind of fault and different fault location. The dip direction is indicated using two selected methods based on analysis of energy and impedance variation during the disturbance. The methods are applied for the measurements aggregated in different nodes of power systems, representing supply and customer side respectively. Based on the results of the dip direction achieved by investigated methods a general conclusions are formulated about the correctness, advantages and disadvantages of the discussed methods.

Keywords— power quality, voltage dip, propagation of voltage dip, voltage dip direction finder

I. INTRODUCTION

It is believed that voltage dip is one of the most prominent problems associated with the power quality. It is especially dangerous for customers who are sensitive to supply voltage disturbances, leads to downtime of industrial factories, which generates losses of industrial factories. The main cause of voltage dips are short circuits in the power system. These faults are not limited to the location of origin of the phenomena but are also transmitted by impedance of the power system. That is why it is important to identify what caused the voltage dip and where is the location of the phenomena. The location process in rough approach might be limited to determination of the direction of the dip which means identification of on which side of the measuring point the voltage dip has occurred. It is strategic in terms of responsibility for the consequence of the disturbance. A full location process usually consists of two steps, the first is information whether the voltage dip occurred "upstream" or "downstream" of the measuring point, i.e. it leads to information if the dip is located on the energy customer or distribution system operator side. Then a separated algorithms are used to determine the place where the voltage dip occurs.

Several methods have been described in the literature to help in detecting the sources of voltage dip. The methods are based mainly on: the analysis of voltage and current waveforms [1],[2],[3], the analysis of the system operation trajectory during the dip, [5],[11], the analysis of the equivalent electric circuit, [3], the analysis of power and energy during the disturbance [7], [13], the analysis of voltages [3], asymmetry factor value and symmetric component phase angle [3],[8],[9], vector-space approach [7] and algorithms for the operation of protection automatics systems, they are composed of impedance variation analysis, the analysis of current real part and "distance" protection [4],[8],[10].

This paper focuses on analyzing of the method based on the algorithms which are used in power system protection techniques taking into account impedance variation analysis and a method based on the analysis of energy during the disturbance. The study model was used to simulate the faults in low voltage networks, in which carried a short circuit along a power supply line and in the customer side. Measurements were made at the same time at the transformer low voltage terminal (the supply side) and at the connection point of the customer side. Simultaneous measurements of voltages and currents on the supplier and customer sides allowed to validate the operation of the investigated methods of the voltage dip direction finder, regardless of the measuring point.

II. CONCEPT OF THE PROPOSED METHODS

A. Basic idea of impedance variation analysis

The main reason of voltage dips are short circuit occurring in power systems, therefore for location of the source of the voltage dip the information related to the operation of power system protection systems is used. The authors of the papers [4],[8],[10] discussed in detail the methods related to the systems of automatic and protection.

One of the method is based on the impedance variation during voltage dip. Detailed theoretical analysis was presented in the literature [10]. The diagram of Fig. 1 illustrates the idea of the location of the voltage dip source which is represented by the equivalent single-phase circuit. $\underline{Z}_1$ and $\underline{E}_1$ are equivalent impedance and an internal voltage source on the supply side. While $\underline{Z}_2$ and $\underline{E}_2$ are the parameters of the circuit are linear and determined for the fundamental component. In the point of the connection the measurements of voltage and current are located. In mentioned method it is proposed to calculate transient impedance which is calculated on the basis of changes in the voltage and current during the disturbances. The changes of the impedances can be presented in the complex plane (Fig. 2.). Depending on in which quarter of the complex plane the transition of the impedance is located, the voltage dip direction can be determined.

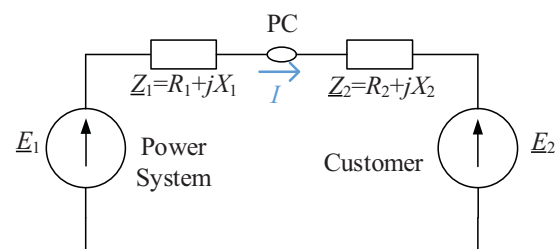


Fig. 1. Equivalent circuit for impedance analysis during voltage dip.

In the impedance variation method, based on the measured currents and voltages at the point of the connection, the impedance can be determined according to the following equation:

$$\underline{Z_e} = \frac{\Delta \underline{U}}{\Delta \underline{I}} = \frac{\underline{U}_{\text{voltage dip}} - \underline{U}_{\text{pre voltage dip}}}{\underline{I}_{\text{voltage dip}} - \underline{I}_{\text{pre voltage dip}}} \quad (1)$$

where: $\underline{U}_{\text{voltage dip}}$, $\underline{I}_{\text{voltage dip}}$, $\underline{U}_{\text{pre voltage dip}}$, $\underline{I}_{\text{pre voltage dip}}$ are pairs of pre-dip and during-dip voltage and current

For short-circuits on the customer side the complex impedance value is located in the first quadrant, while for disturbances on the power system side, the complex impedance value is located in the third quarter. (Fig. 2). It can be stated that if $\text{Real}(\underline{Z_e}) > 0$ the source of sag is on supply side and if $\text{Real}(\underline{Z_e}) < 0$ the source of sag is on customer-side.

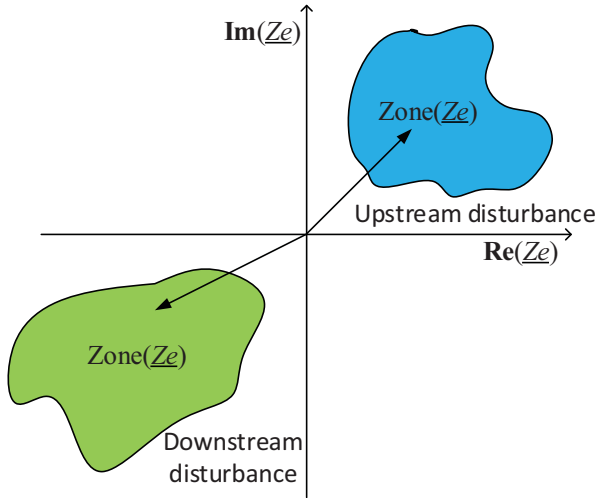


Fig. 2. Illustration of a method based on impedance analysis during voltage dip.

B. Basic analysis of energy during the disturbance

Another method tested in the work is an algorithm based on the sign of disturbance energy. The authors of work [13] described in detail the principle of the mentioned method, which is calculated on the basis of changes in instantaneous power during a disturbance, determined as the difference of disturbance power and steady state power according to the formula:

$$DP(t) = (p(t)_{\text{sag}} - p(t)_{\text{presag}}) \quad (2)$$

where: $p(t) = u(t) \times i(t)$, $u(t)$ and $i(t)$ are phase voltage and current.

From equation (2) it follows that the disturbance power is zero in the steady state while is different than zero in the transient state. Using integral of the power during the time of disturbance the change in the disturbance energy is calculated according to the following equation:

$$DE(t) = \int_0^t DP(\tau) d\tau \quad (3)$$

The negative direction of the energy during the disturbance indicates that the location of the voltage dip source is on the upper side of the measuring instrument, i.e. on the supply side. The assumption for this method is the initial condition based on knowledge of the direction of active energy flow.

III. MODEL AND SIMULATION RESEARCH

Fig. 3 shows a schematic diagram of the MV/LV power system model used for generating voltage and current signals during voltage dips. This model was developed in Matlab/Simulink environment. MV power system $U_{NQ} = 20$ kV is represented as a source of sinusoidal voltage and the equivalent impedance resulting from the assumed short-circuit power. It was assumed that short-circuit power of the power system is $S''_{NQ} = 340$ MVA ratio $Z_0/Z_1 = 1,5$ and $X_1/R_1 = 30$. It was applied 1km cable line of the cross sectional area 240mm² (marked as A). Transformer MV/LV has the following parameters: $\vartheta = 20/0,4$ kV/kV, $S_{NT} = 630$ kVA, short circuit voltage $\Delta U_{k\%} = 6\%$, losses in the load $\Delta P_{Cu} = 6.3$ kW. It was considered Dy5 transformer connection group. The following low-voltage overhead line lengths are assumed: two sections of 250m each with a section of 70mm² (marked as B, C) and a 100m cable line with a 35mm² cross section (designated as D). The unit parameters of the MV overhead line and the LV overhead line and cable are shown in Table I, Table II and Table III.

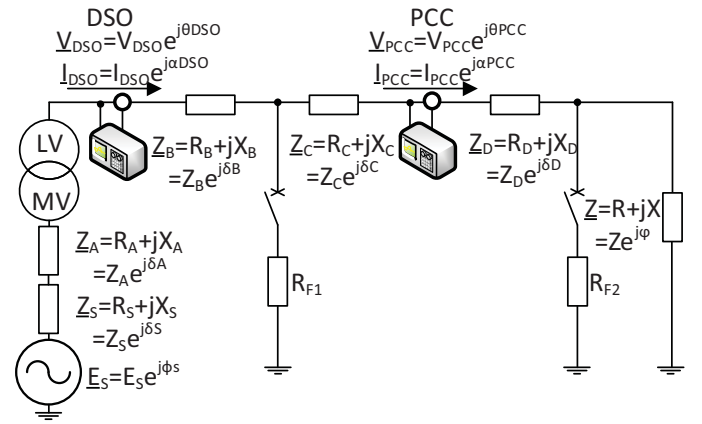


Fig. 3. A simplified model of the power system implemented for analyzing the propagation of voltage dips in LV network.

TABLE I. PARAMETERS OF MODELLED OVERHEAD MV LINE [14],[15]

	R' [Ω/km]	X' [Ω/km]	C' [nF/km]	L_A [km]
Zero sequence component	0,31	0,4	312	1
Positive sequence component	0,8	0,11	520	

TABLE II. PARAMETERS OF MODELLED OVERHEAD LV LINE [14],[15]

	R' [Ω/km]	X' [Ω/km]	C' [nF/km]	l_B [km]	l_C [km]
Zero sequence component	0,58	0,9	90	0,25	0,25
Positive sequence component	0,43	0,25	15		

TABLE III. PARAMETERS OF MODELLED LV CABLE [14],[15]

	R' [Ω/km]	X' [Ω/km]	C' [nF/km]	l_0 [km]
Zero sequence component	2,12	0,29	650	0,1
Positive sequence component	0,53	0,08	108	

As a source of voltage dips it were used the simulations of faults in the middle length of the LV overhead supply line and at the customer load respectively. The measurements were carried out simultaneously at the point of connection of the customer LV (designated as PCC) and on the busbar of LV electrical station (denoted as DSO). The simulations were performed with the fault resistance $R_{F1} = 0.1\Omega$ and $R_{F2} \approx 0\Omega$. It was assumed that simulated faults were transient and has occurred between 0.1 and 0.16s. The aim of the investigations was to implement mentioned voltage dip direction finder methods in both points of measurements, DSO and PCC respectively, and find out if the method can be used in the dips direction process, regardless of the point of observation.

A. Fault in the supply line

Fig. 4 shows the case of a three-phase fault on the customer side, i.e. representing the case when the fault was simulated at the end of the LV cable inside of the customer installation, at the customer load. From the point of view of the measuring systems located on the transformer busbars supplying the LV network (DSO side) and the point of connection of the customer (PCC side), the discussed case corresponds to the direction of the voltage dip point located downstream from the measuring instruments.

Fig. 4a) and 4c) illustrates voltage and current waveforms at the DSO point, i.e. on the secondary side of the MV / LV supply transformer. Fig. 4b) and 4d) show the voltage and current waveforms at the PCC measuring point. Fig. 4e) presents the changes of the real part of the impedance seen at the DSO point. During distortion observed in DSO point changes of real part of the impedance is less than zero, which means that the voltage dip is located downstream. Fig. 4f) shows changes of the real part of the impedance observed at the PCC measuring point. It can be noticed that in PCC point of measurement the observed changes of real part of the impedance is also less than zero. Thus it can be summarize that the methods indicates that voltage dip phenomena located downstream of the PCC. Fig. 4g) shows the results of the disturbance energy waveform at the DSO point of measurement while Fig. 4h) shows the disturbance energy waveform at the PCC point. In both measurements points calculated energy disturbance is greater than zero, which means that the location of the source of disturbance is downstream side of the measuring points. In presented example it can be concluded that both methods, impedance variation analysis and energy during the disturbance, correctly indicate the direction of the voltage dip location.

Fig. 5 shows a case of a single-phase to ground fault at half of LV overhead supplied line. From the point of view of the measuring systems located on the transformer busbars, supplying the LV network (DSO), the discussed case

corresponds to the location of the source of the disturbance downstream of the measuring point, while from the point of view of measurements located at the point of connection of the customers (PCC) the discussed case corresponds to the direction of the voltage dip upstream of the measurements point.

B. Fault at the customer load

Fig. 5a) and 5c) illustrate voltage and current waveforms in the DSO point, while Fig. 5b) and 5d) show voltage and current waveforms at the PCC measuring point. Fig. 5e) shows changes of the real part of the impedance at the point of DSO. The value of impedance during the transient is less than zero, which means that the location of the sources of disturbances is downstream of the measuring point in DSO, i.e. the source of the phenomena is inside the LV network. Fig. 5f) shows the real part of the impedance observed at the customer PCC measurements point. It can be seen that in this case the real part of the impedance during the disturbance obtain positive values, which indicates that the location of the voltage dip source is upstream of the PCC measurement point. Fig. 5g) shows the waveform distortion energy at the point of DSO. Obtained change of the disturbance energy is greater than zero, which means that the location of the disturbance source is downstream of the DSO measurement point. Fig. 5h) shows the disturbance energy waveform at the PCC point. For this point of measurements the calculated disturbance energy is less than zero, which allows to conclude that the location of the disturbance is upstream of the PCC measurement point, i.e. on the supply side. In the case of a investigated single-phase to ground fault scenario located in the supply line, it should be noted that conclusion about the direction of the disturbance is not the same for each phase. The calculated disturbance energy have different sign, positive or negative. The most dominant change decides about the identification of the direction finder. Generally, it can be concluded that using measurements in both points DSO and PCC, the identification of the direction of the voltage dip occurrence using tested methods is correct.

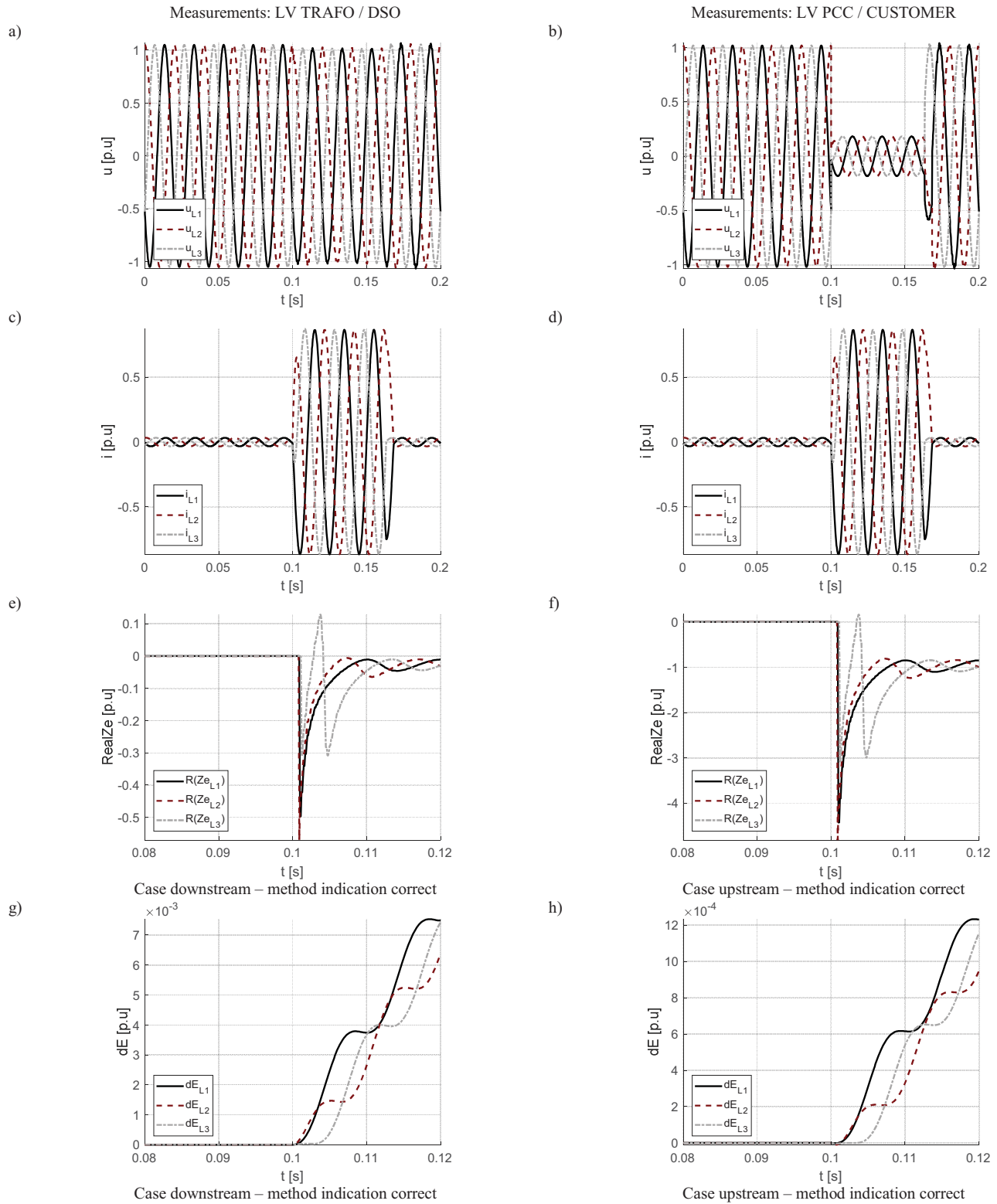


Fig. 4. Case of three phase to ground fault at the customer side (LV load), $R_{F2} \approx 0\Omega$: a) and c) simulated voltage and current waveforms at the DSO point, b) and d) simulated voltage and current waveforms at the PCC, e) the result of changes of the real part of the impedance at the DSO, f) the result of changes of the real part of the impedance at the PCC, g) the result of the disturbance energy waveform at the DSO, h) the result of disturbance energy waveform at the PCC.

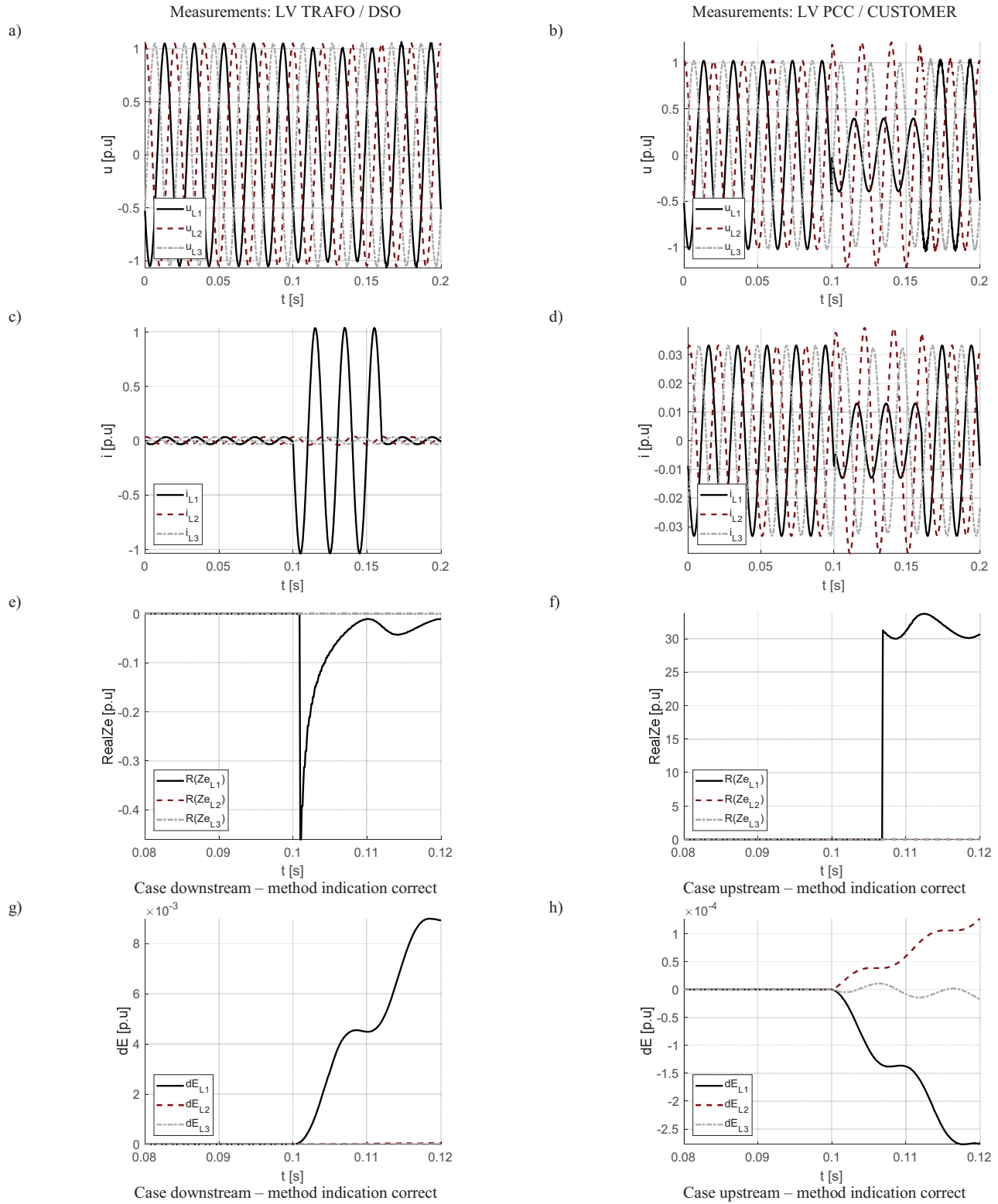


Fig. 5. Case of one phase to ground fault at half of the LV overhead line, $R_{FI} \approx 0.1 \Omega$: a) and c) simulated voltage and current waveforms at the DSO point, b) and d) simulated voltage and current waveforms at the PCC, e) the result of changes of the real part of the impedance at the DSO, f) the result of changes of the real part of the impedance at the PCC, g) the result of the disturbance energy waveform at the DSO, h) the result of disturbance energy waveform at the PCC.

IV. CONCLUSIONS

This work presents results of application of two selected methods of dips direction finder. The first of the discussed methods is the impedance variation analysis during the transient state caused by the dip. Second method is based on the analysis of energy during the disturbance. Both methods have been implemented and tested in Matlab environment on the basis of the simulations of short circuits localized at various points in the supply low voltage power system. The idea of the investigation was to implement mentioned dip direction finder methods in two measurement points related to distribution system operator side, i.e. on the secondary side of the MV / LV transformer, and related to the customer side, i.e. in the point of the connection of the customer. Then the results of the dip direction returned by the investigated methods was examined using faults located in the supply line and at the customer load respectively. Results of the investigations show that investigated methods correctly localize the direction of the voltage dip location in point of view of the measurements points. Particularly, using investigated methods it can be noticed that conclusion about the direction of the dip location is not clear for each phase in case of asymmetrical fault. It is especially evident in case of the method based on energy during disturbance. In order to achieve proper direction finder it is proposed to use dominant impedance or energy change. Studies have confirmed the effectiveness of applying methods regardless of the place of measurement. It means that for presented case of fault located in the supply line a proper direction of the dip was indicated as downstream for the measurements at the supply transformer and upstream for the measurements in the point of the connection of the customer.

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